



Learning Python - Sheet #9

Pygame

Pygame is a python library which makes it easy to provide interactive games with graphics, sounds, ...

1. Pygame

Let's check that pygame is installed. In IDLE Shell, type:

```
import pygame
```

If you get a **ModuleNotFoundError: No module named 'pygame'** message then you need to install pygame. Otherwise, you're good to go.

2. Our first pygame program

In a new file, type the following. You'll need to download a picture of a ball and call it ball.png.

```
import pygame

pygame.init()
screen = pygame.display.set_mode((800, 600))
clock = pygame.time.Clock()
running = True
ball = pygame.image.load('ball.png')
x, y = 0, 0
vx, vy = 10, 10

while running:
    # poll for events
    # pygame.QUIT event means the user clicked X to close your window
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False

    # fill the screen with a color to wipe away anything from last frame
    screen.fill("black")

    # show the ball
    screen.blit(ball, (x, y))

    # move the ball
    x += vx
    y += vy
    if x >= screen.width:
        vx = -vx
    if y >= screen.height:
        vy = -vy

    # flip() the display to put your work on screen
    pygame.display.flip()

    clock.tick(60) # limits FPS to 60

pygame.quit()
```



Pay attention:

The function **enumerate** allows to get both the element and its index in the list. We could write **for element in my_list:** but then we wouldn't know the element's position in the list.



Practice to progress:

Can you extend the program to take control of the ball?

Tip: You can know that a key is pressed from **pygame.KEYDOWN** event and you can find which key it is by looking at **event.key**. For example, **pygame.UP**, **pygame.DOWN**, **pygame.LEFT** and **pygame.RIGHT** for arrows.

Or use **pygame.key.get_pressed()** to get a list of key states. Index it with key codes like **pygame.UP**

